Lab Documentation: Responsive Web Design – "Design a Piano"

Project Overview: Built a responsive, interactive piano keyboard UI using semantic HTML5 and CSS3 as part of the freeCodeCamp Responsive Web Design Certification.

Technical Implementation & Workflow

1. Structural Markup (`index.html`)

- Defined standard document metadata and linked external stylesheet (`styles.css`).
- Constructed main wrapper elements: `#piano` container and `.keys` layout.
- Structured keys into functional groups across octaves using repeating combinations of white (`.key`) and black (`.key.black--key`) piano keys.
- Embedded logo element within the piano header.

2. Styling & Layout Engine (`styles.css`)

- **Box Model Baseline:** Applied global `box-sizing: border-box` inheritance rules.
- **Layout Positioning:**
  - Positioned the outer `#piano` container (`position: relative`) with custom background, padding, and centered margins.
  - Used `float: left` and controlled dimensions (`width: 41px`, `height: 175px`) on `.key` elements for sequential horizontal alignment of white keys.
- **Pseudo-Elements for Overlay:**
  - Implemented `.key.black--key::after` pseudo-elements (`position: absolute`, `left: -18px`) to visually overlap black keys between adjacent white keys without disrupting the document flow.
- **Responsive Styling:**
  - Configured `@media (max-width: 768px)` media query to adjust the `#piano` container width down to `358px` for mobile viewports.
  - Applied `position: absolute` top positioning to pin the logo element within the frame.

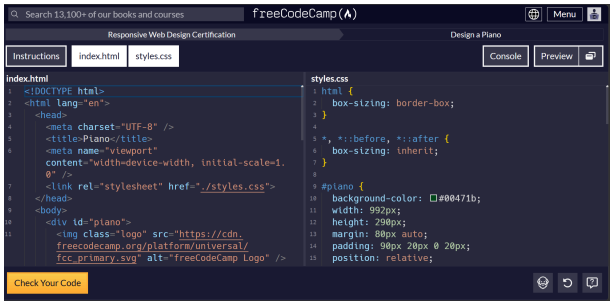
Core Knowledge Gained

- **CSS Pseudo-Elements (`::after`):** Applied absolute positioning on pseudo-elements to create layered complex UI components without adding bloated DOM nodes.

- **Layout Flow Control:** Combined floating elements (float: left) with overflow clearing (overflow: hidden) to maintain structured alignment across child containers.

- **Responsive Adaptability:** Implemented media queries and relative pixel boundaries to render fluid UI layouts across desktop and mobile screens.

CODE:









PREVIEW:

